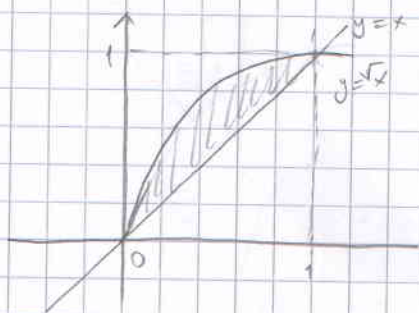


AREA BETWEEN CURVES

- ① Find the area of the region bounded by the curves $y = \sqrt{x}$ and $y = x$



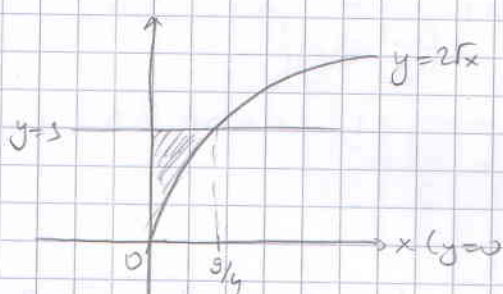
$$\sqrt{x} = x \Rightarrow x = x^2 \Rightarrow x^1 - x = 0 \quad x(x-1) = 0$$

$$x_1 = 0 \quad x_2 = 1$$

$$A = \int_0^1 (\sqrt{x} - x) dx = \left(\frac{2}{3} x^{3/2} - \frac{x^2}{2} \right) \Big|_0^1$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \text{ br}^2 //$$

- ② Find the area of the region bounded by the curve $y^2 = 4x$ and the lines $y = 3$ and $x = 0$

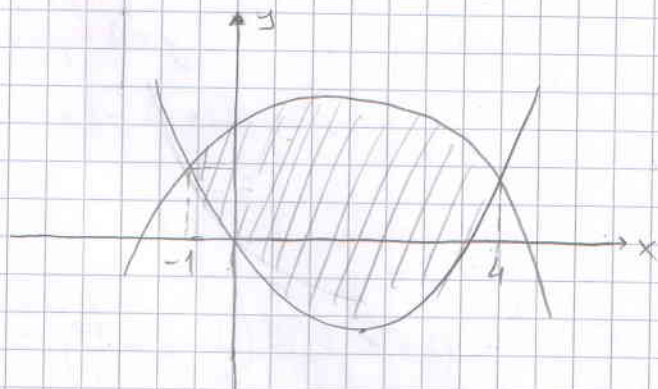


$$\text{I. Vol.} \int_0^{9/4} (3 - 2\sqrt{x}) dx = \left(3x - \frac{4}{3} x^{3/2} \right) \Big|_0^{9/4}$$

$$= \frac{27}{4} - \frac{4}{3} \cdot \frac{27}{8} = \frac{9}{4} \text{ br}^2 //$$

$$\text{II. Vol.} \int_0^3 \left(\frac{y^2}{4} - 0 \right) dy = \frac{y^3}{12} \Big|_0^3 = \frac{27}{12} = \frac{9}{4} \text{ br}^2 //$$

- ③ Find the area of the region bounded by the curves $y = 4x - x^2 + 8$ and $y = x^2 - 2x$



$$-x^2 + 4x + 8 = x^2 - 2x$$

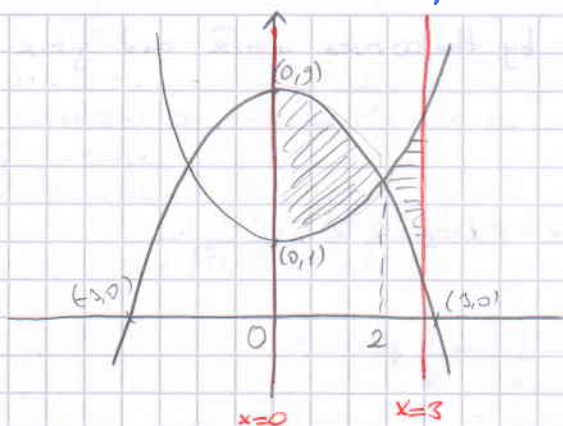
$$2x^2 - 6x - 8 = 0 \Rightarrow x^2 - 3x - 4 = 0$$

$$x = -1 \quad x = 4$$

$$\text{Area} = \int_{-1}^4 (-x^2 + 4x + 8 - x^2 + 2x) dx = \int_{-1}^4 (-2x^2 + 6x + 8) dx = \left(-\frac{2}{3} x^3 + 3x^2 + 8x \right) \Big|_{-1}^4$$

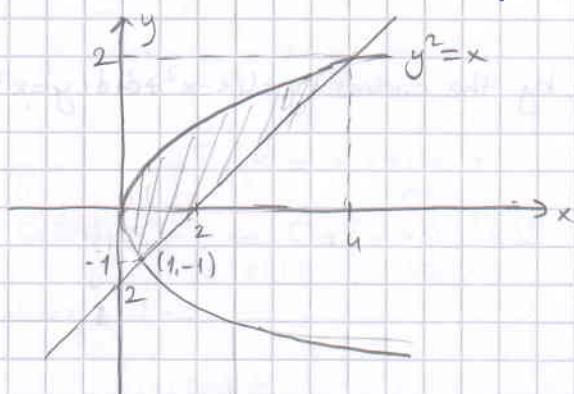
$$= -\frac{128}{3} + 48 + 32 - \frac{2}{3} - 3 + 8 = 85 - \frac{130}{3} = \frac{125}{3} \text{ br}^2 //$$

- ④ Find the area of the region between the curves $y=9-x^2$ and $y=x^2+1$ from $x=0$ to $x=3$



$$\begin{aligned}
 \text{Area} &= \int_0^2 [(9-x^2) - (x^2+1)] dx + \int_2^3 [(x^2+1) - (9-x^2)] dx \\
 &= \int_0^2 (8-2x^2) dx + \int_2^3 (2x^2-8) dx = \left(8x - \frac{2x^3}{3}\right) \Big|_0^2 + \left(\frac{2x^3}{3} - 8x\right) \Big|_2^3 \\
 &= 16 - \frac{16}{3} + \left(18 - 24 - \frac{16}{3} + 16\right) = 26 - \frac{32}{3} = \frac{46}{3} \text{ br}^2
 \end{aligned}$$

- ⑤ Find the area of the region bounded by the graphs $y^2=x$ and $x-y=2$



Intersect. points: $y^2-y=2 \Rightarrow y^2-y-2=0$
 $y=2 \quad y=-1$

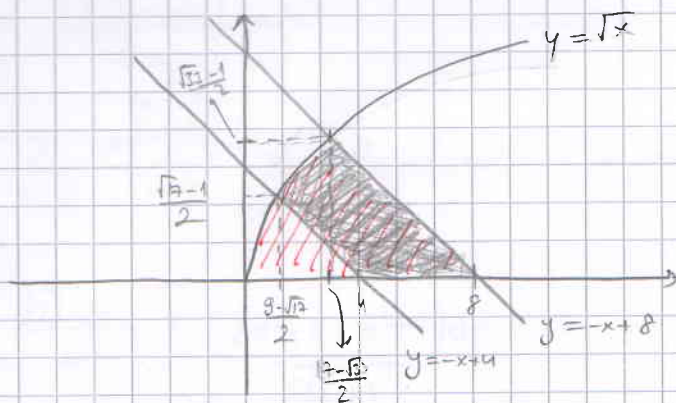
$$\begin{aligned}
 \text{Area} &= \int_{-1}^2 (y+2-y^2) dy = \left(\frac{y^2}{2} + 2y - \frac{y^3}{3}\right) \Big|_{-1}^2 = \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right) \\
 &= 8 - \frac{1}{2} - \frac{9}{3} = 5 - \frac{1}{2} = \frac{9}{2} \text{ br}^2
 \end{aligned}$$

II-Val:

$$\text{Area} = \int_0^1 (\sqrt{x} - (-\sqrt{x})) dx + \int_1^4 (\sqrt{x} - (x-2)) dx =$$

⑥ Find the area of the region bounded by the graphs of $y=\sqrt{x}$ and $y=0$, $y=-x+4$

and $y=-x+8$



$$-x+4=\sqrt{x} \Rightarrow x^2-8x+16=x$$

$$x^2-9x+16=0$$

$$\Delta=17 \quad x=\frac{9-\sqrt{17}}{2}$$

$$-x+8=\sqrt{x} \Rightarrow x^2-16x+64=x$$

$$x^2-17x+64=0$$

$$\Delta=33 \quad x=\frac{17-\sqrt{33}}{2}$$

Area = Red Area - Free Area

$$\text{Red Area} = \int_0^{\frac{9-\sqrt{17}}{2}} \sqrt{x} dx + \int_{\frac{9-\sqrt{17}}{2}}^8 (-x+8) dx = \frac{2}{3} x^{3/2} \Big|_0^{\frac{9-\sqrt{17}}{2}} + \left(-\frac{x^2}{2} + 8x \right) \Big|_{\frac{9-\sqrt{17}}{2}}^8$$

$$\text{Free Area} = \int_0^{\frac{9-\sqrt{17}}{2}} \sqrt{x} dx + \int_{\frac{9-\sqrt{17}}{2}}^4 (-x+4) dx = \frac{2}{3} x^{3/2} \Big|_0^{\frac{9-\sqrt{17}}{2}} + \left(-\frac{x^2}{2} + 4x \right) \Big|_{\frac{9-\sqrt{17}}{2}}^4$$

II. Sol:

$$\text{Area} = D_1 + D_2 + D_3 =$$

III. Sol:

$$\text{Area} = \int_0^{\frac{\sqrt{17}-1}{2}} ((8-y)-(4-y)) dy + \int_{\frac{\sqrt{17}-1}{2}}^{\frac{\sqrt{17}-1}{2}} (8-y-y^2) dy$$

Leibnitz Formula

⑦ $F(x) = \int_{1-x}^{2-x} e^{t^2} dt$ $F'(1) = ?$

$f(x, t) = t^2 \Rightarrow \frac{\partial f}{\partial x} = 0$

$F'(x) = \int_{1-x}^{2-x} 0 \cdot dt + (2-x) \cdot e^{(2-x)^2} - e^{(1-x)^2} (1-x)' = e^{(1-x)^2} - e^{(2-x)^2}$

$F'(1) = e^{(1-1)^2} - e^{(2-1)^2} = 1 - e$

⑧ $\lim_{x \rightarrow 0} \frac{\arctan x}{\int_0^{2x} e^{t^2} dt}$

$\lim_{x \rightarrow 0} \frac{\arctan x}{\int_0^{2x} e^{t^2} dt} = \lim_{x \rightarrow 0} \frac{\frac{1}{1+x^2}}{2 \cdot e^{(2x)^2} - 0} = \lim_{x \rightarrow 0} \frac{1}{2(1+x^2)e^{(2x)^2}} = \frac{1}{2}$

⑨ $F(t) = \int_1^t \frac{\sin(x+t)}{x} dx$ $F'(t) = ?$

$f(t, x) = \frac{\sin(x+t)}{x} \Rightarrow \frac{\partial f}{\partial t} = \frac{x \cdot \cos(x+t)}{x} = \cos(x+t)$

$F'(t) = \int_1^t \cos(x+t) dx + 1 \cdot \frac{\sin t^2}{t} - 0 \cdot \frac{\sin(1+t)}{1} = \frac{\sin(x+t)}{t} \Big|_1^t + \frac{\sin t^2}{t}$
 $= \frac{\sin t^2}{t} - \frac{\sin t}{t} + \frac{\sin t^2}{t} = \frac{2 \sin(t^2) - \sin t}{t}$

⑩ $F(x) = \int_0^{\sqrt{x}} e^{t^2} dt$ $F'(x) = ?$

$\frac{\partial(e^{t^2})}{\partial x} = 0 \Rightarrow F'(x) = 0 + \frac{1}{2\sqrt{x}} \cdot e^x - 0 \cdot e^0 = \frac{e^x}{2\sqrt{x}}$

$$(11) \lim_{x \rightarrow \pi} \frac{\int_{\pi}^x \frac{dt}{2+\cos t}}{x-\pi}$$

$$\frac{D\left(\frac{1}{2+\cos t}\right)}{Dx} = 0$$

$$\lim_{x \rightarrow \pi} \frac{\int_{\pi}^x \frac{dt}{2+\cos t}}{x-\pi} = \lim_{x \rightarrow \pi} \left(0 + 1 \cdot \frac{1}{2+\cos x} - 0 \cdot \frac{1}{2+\cos \pi}\right) = \lim_{x \rightarrow \pi} \frac{1}{2+\cos x} = \frac{1}{2-1} = 1$$

$$(12) \frac{d}{dt} \int_1^{4t} \underbrace{(\sqrt{x^2+1} + t^2+1)}_{f(x,t)} dx$$

$$\frac{Df(x,t)}{Dt} = 2t$$

$$\frac{dF(t)}{dt} = \int_1^{4t} 2t dx + 0 \cdot (\sqrt{4^2+1}) - 0 \cdot (\sqrt{1^2+1}) = 2t \cdot x \Big|_1^{4t} = 8t - 2t = 6t$$

$$(13) F(x) = \frac{1}{x} \int_1^x [2t - F'(t)] dt \text{ is given } F'(1) = ?$$

$$F'(x) = \frac{-1}{x^2} \int_1^x [2t - F'(t)] dt + \frac{1}{x} \left[\int_1^x 0 dt + 1 \cdot (2x - F'(x)) - 0 \cdot (2 - F'(1)) \right]$$

$$F(x) = \frac{-1}{x^2} (t^2 - F(t)) \Big|_1^x + 2 - \frac{F'(x)}{x}$$

$$F'(1) = \frac{-1}{1} \left(t^2 - F(t) \right) \Big|_1^1 + 2 - F'(1) \Rightarrow 2F'(1) = 2 \Rightarrow F'(1) = 1$$

II. Sol: $\int_1^x (2t - F'(t)) dt = t^2 - F(t) \Big|_1^x = x^2 - F(x) - (1 - F(1))$

$$F(x) = x - \frac{F(x)}{x} + \frac{F(1) - 1}{x}$$

$$F'(x) = 1 - \frac{F(x) \cdot x - F(x) \cdot 1}{x^2} - \frac{F(1) - 1}{x^2}$$

$$F'(1) = 1 - \frac{F(1) \cdot 1 - F(1)}{1} - \frac{F(1) - 1}{1} = 1 - F'(1) + F(1) - F(1) + 1$$

$$2F'(1) = 2 \Rightarrow F'(1) = 1$$

APPROXIMATE INTEGRATION

① Use the trapezoidal rule with $n=5$ to approximate the integral $\int_1^2 \frac{1}{x} dx$

$$\Delta x = \frac{2-1}{5} = \frac{1}{5} = 0,2 \quad f(x) = \frac{1}{x}$$

$$\int_1^2 \frac{dx}{x} \approx \frac{0,2}{2} \cdot [f(1) + 2f(1,2) + f(1,4) + 2f(1,6) + f(1,8) + f(2)]$$

$$\approx 0,1 \cdot \left[\frac{1}{1} + 2 \cdot \frac{1}{1,2} + 2 \cdot \frac{1}{1,4} + 2 \cdot \frac{1}{1,6} + 2 \cdot \frac{1}{1,8} + \frac{1}{2} \right] \approx 0.695635$$

$$\int_1^2 \frac{dx}{x} = 0.6931 \quad \text{but} \quad \int_1^2 \frac{dx}{x} \approx 0.695635$$

② Use the Simpson's Rule with $n=10$ to approximate the integral $\int_1^2 \frac{1}{x} dx$

$$\int_1^2 \frac{dx}{x} = \frac{\Delta x}{3} [f(1) + 4f(1,1) + 2f(1,2) + 4f(1,3) + 2f(1,4) + 4f(1,5) + 2f(1,6) + 4f(1,7) + 2f(1,8) + 4f(1,9) + f(2)]$$

$$= \frac{0,1}{3} \left(\frac{1}{1} + 4 \cdot \frac{1}{1,1} + 2 \cdot \frac{1}{1,2} + 4 \cdot \frac{1}{1,3} + 2 \cdot \frac{1}{1,4} + 4 \cdot \frac{1}{1,5} + 2 \cdot \frac{1}{1,6} + 4 \cdot \frac{1}{1,7} + 2 \cdot \frac{1}{1,8} + 4 \cdot \frac{1}{1,9} + \frac{1}{2} \right)$$

$$= 0,693150$$